INCH-POUND
MIL-DTL-16377/58C(SH)
26 May 2023
SUPERSEDING
MIL-F-16377/58B(SH)
11 April 1983

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; FLUORESCENT, DETAIL LIGHTING FOR MIRRORS, 120 VOLTS, 60 HERTZ, DRIPPROOF, SYMBOLS 351, 351.1, 352, 352.1, 351SSL, 351.1SSL, 352SSL, AND 352.1SSL

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

Unless otherwise noted, all dimensions are in inches. For tolerances other than those specified herein, see 3.5.2 of MIL-DTL-16377.

Figures 1 through 8 provide information for the light fixtures covered by this specification sheet. See the requirements sections for specific references to these figures.

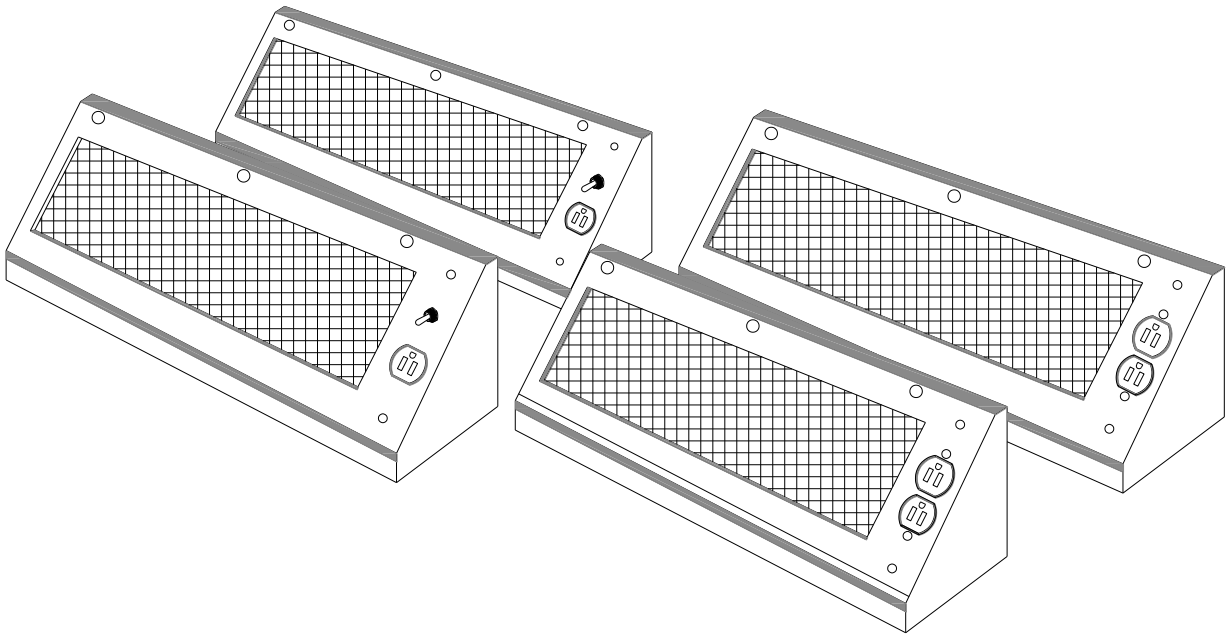


FIGURE 1. Typical view of mirror fixtures.

AMSC N/A FSC 6210

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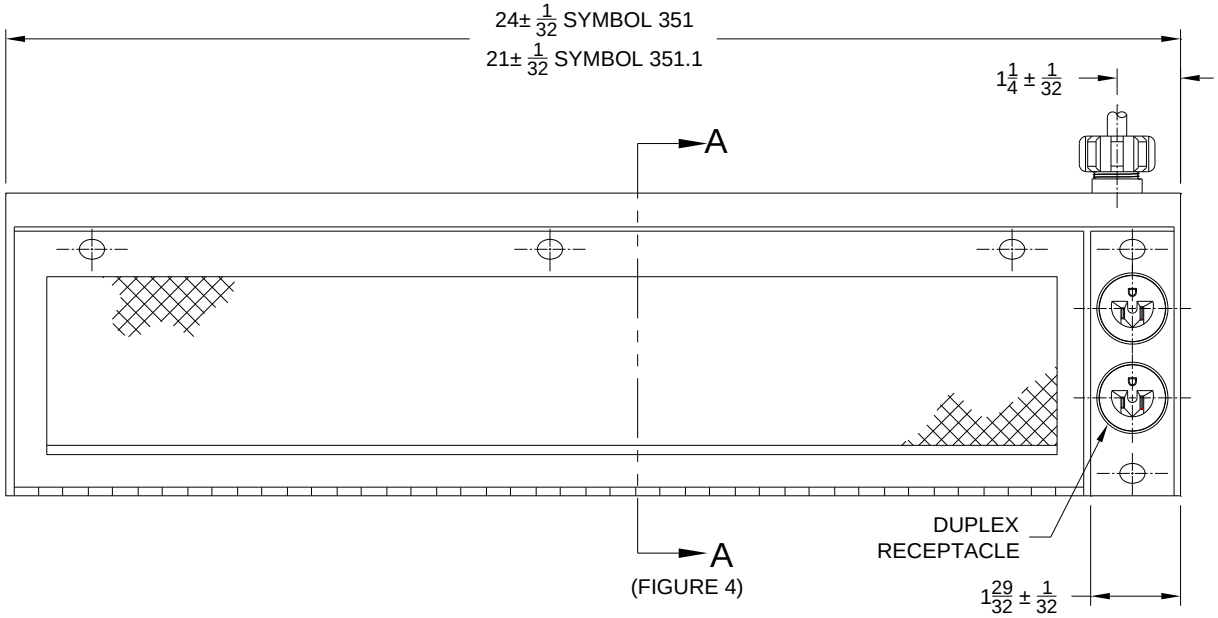


FIGURE 2. Typical front view of symbols 351 and 351.1.

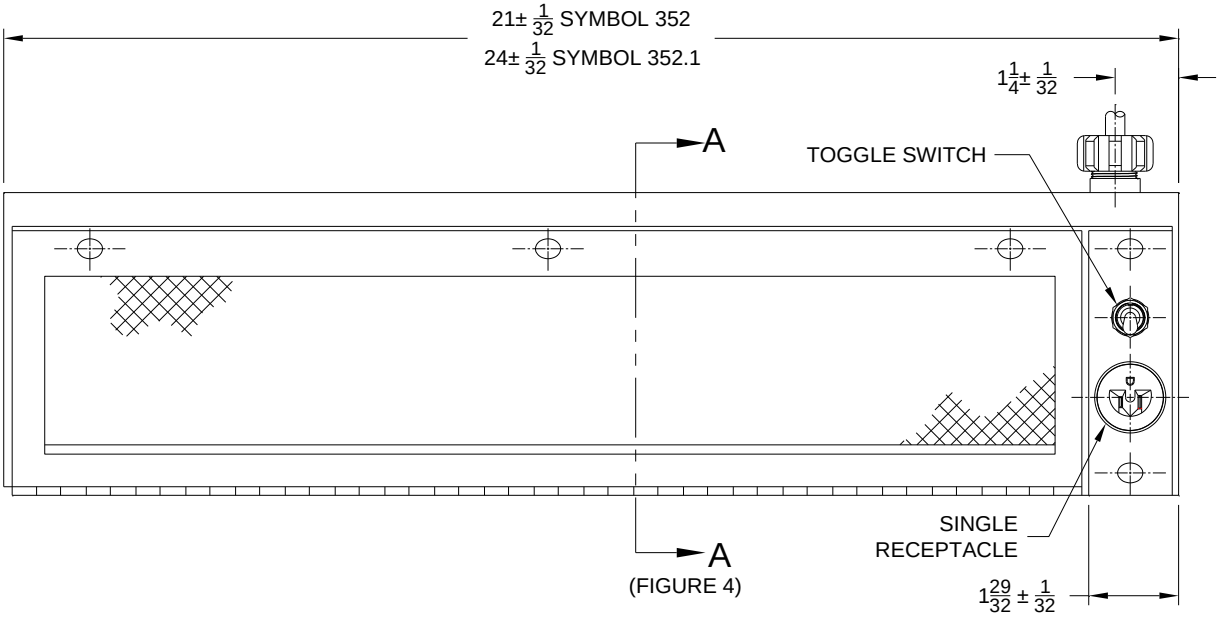
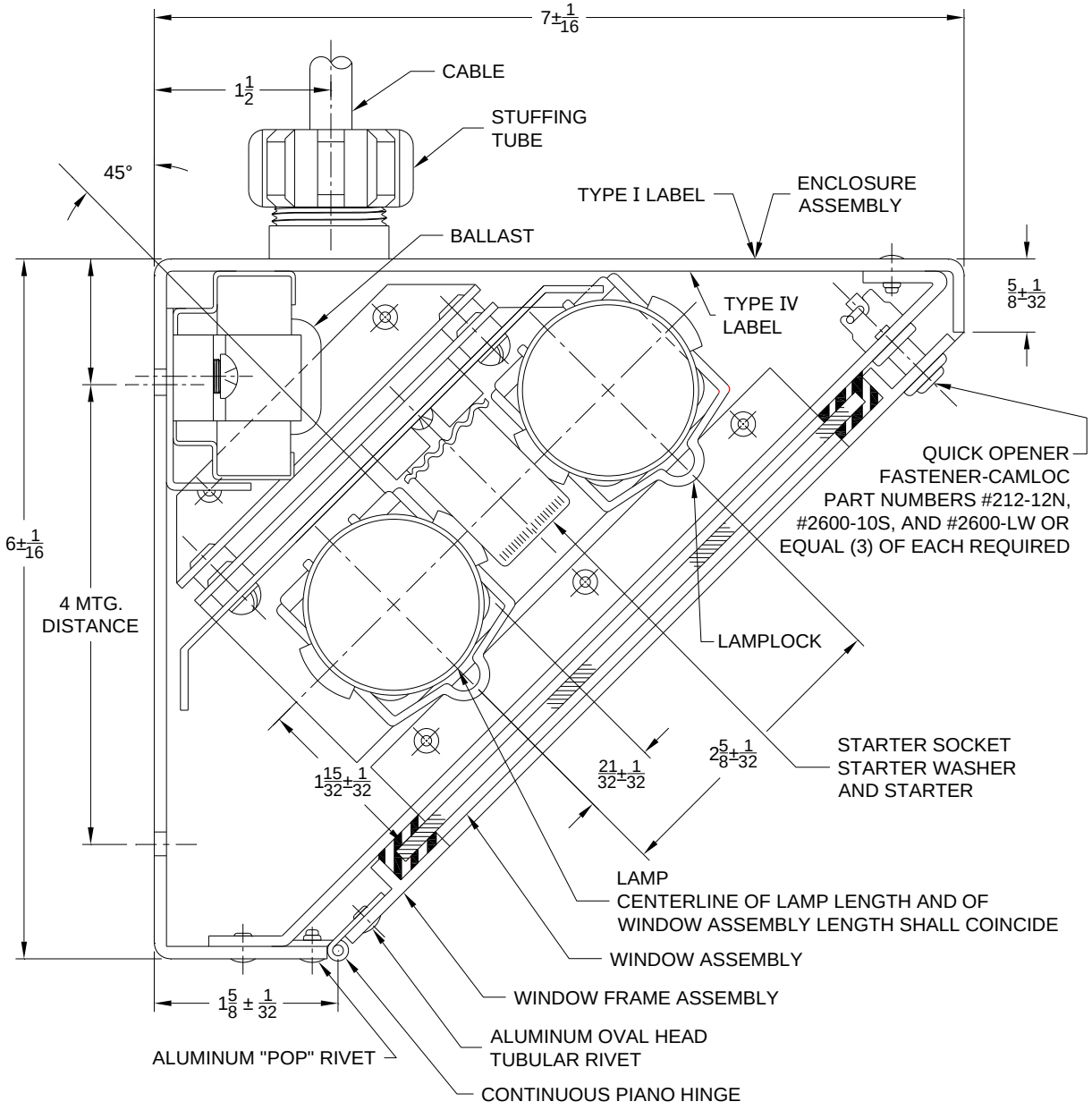


FIGURE 3. Typical front view of symbols 352 and 352.1.

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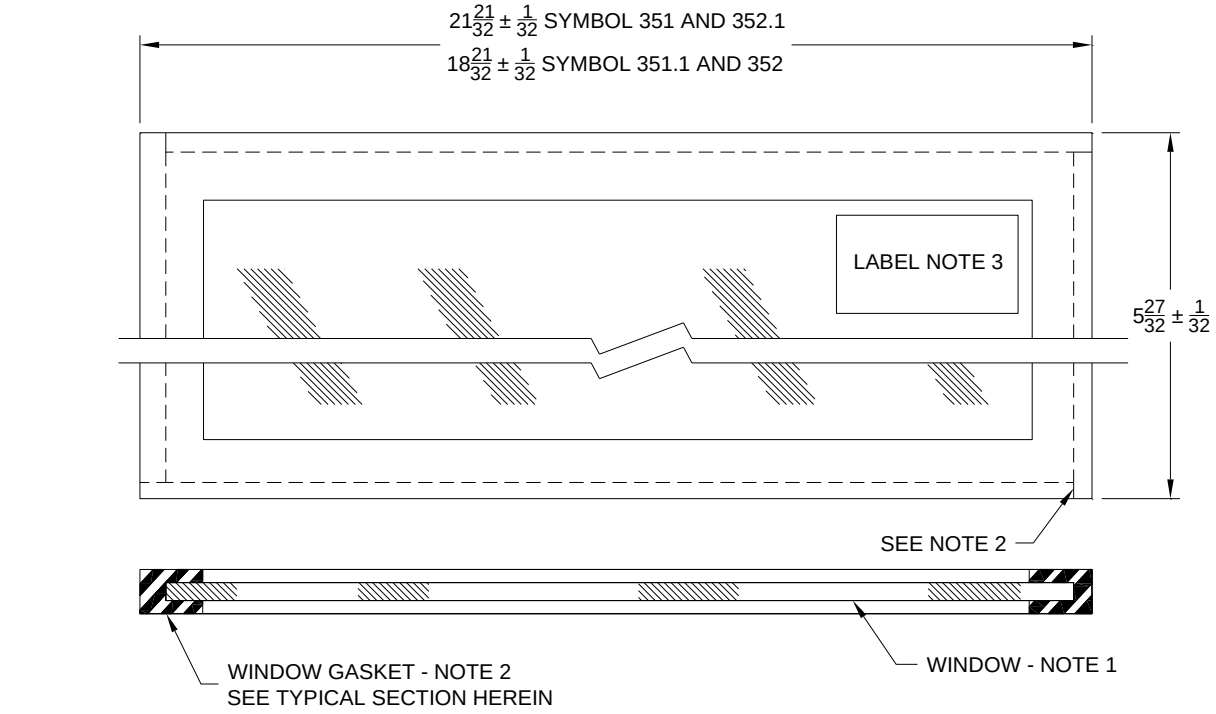


SECTION A-A OF FIGURES 2 AND 3

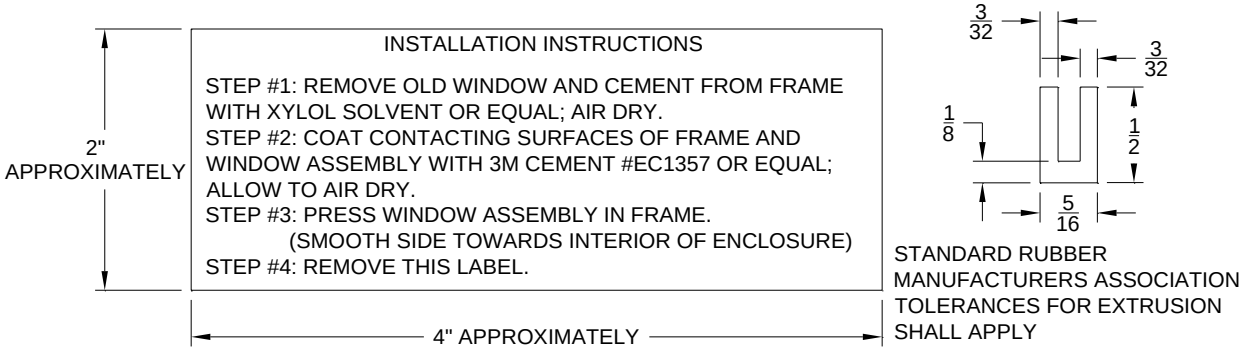
NOTE: Dimensions shown may be modified to accommodate ballast. Critical dimensions (mounting, window area, and lamp spacing) shall be maintained.

FIGURE 4. Typical section view.

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(A) WINDOW ASSEMBLY



(B) LABEL - NOTE 3

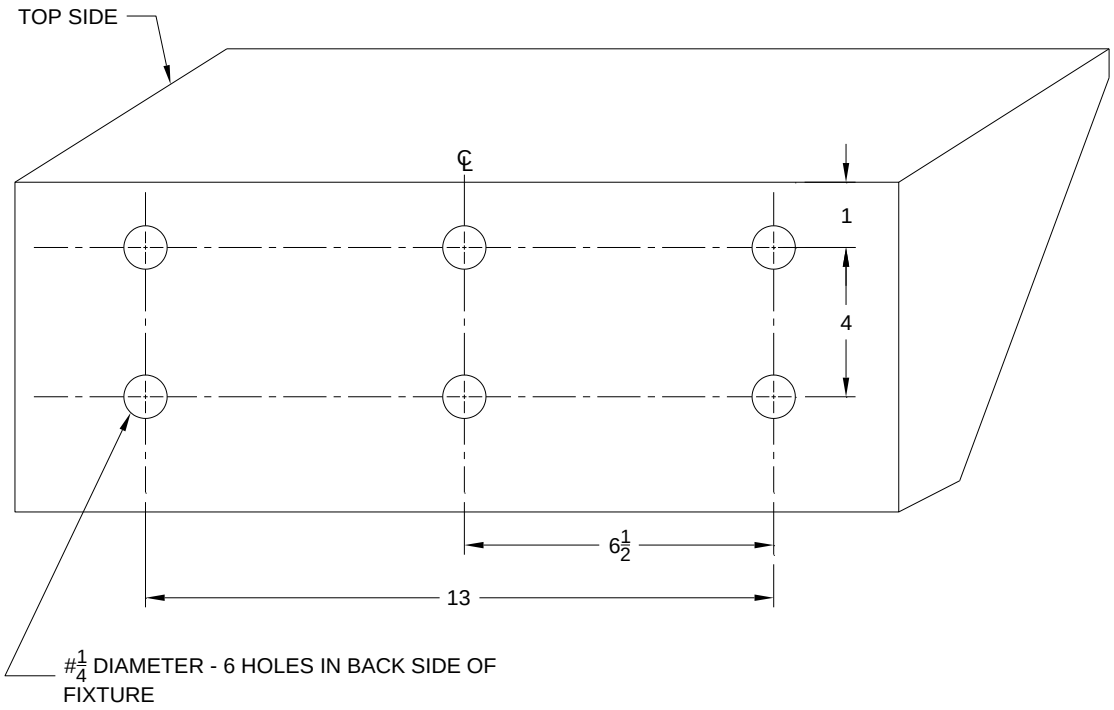
(C) GASKET - TYPICAL SECTION

NOTES:

1. Material for window shall be ⅛-inch thick, clear, prismatic (one side only), heat resistant, cast acrylic plastic sheet, MIL-DTL-24191, type CLEAR-2.
2. Material for window gasket shall be neoprene rubber, 25-35 durometer. Gasket shall be cemented to window. Gasket may be furnished in segments.
3. For window assemblies furnished as repair parts, a type IV label shall be furnished in accordance with MIL-DTL-16377. Location and the information to be shown on this label shall be as shown hereon.
4. Window assembly shall be cemented to window frame.

FIGURE 5. Window assembly.

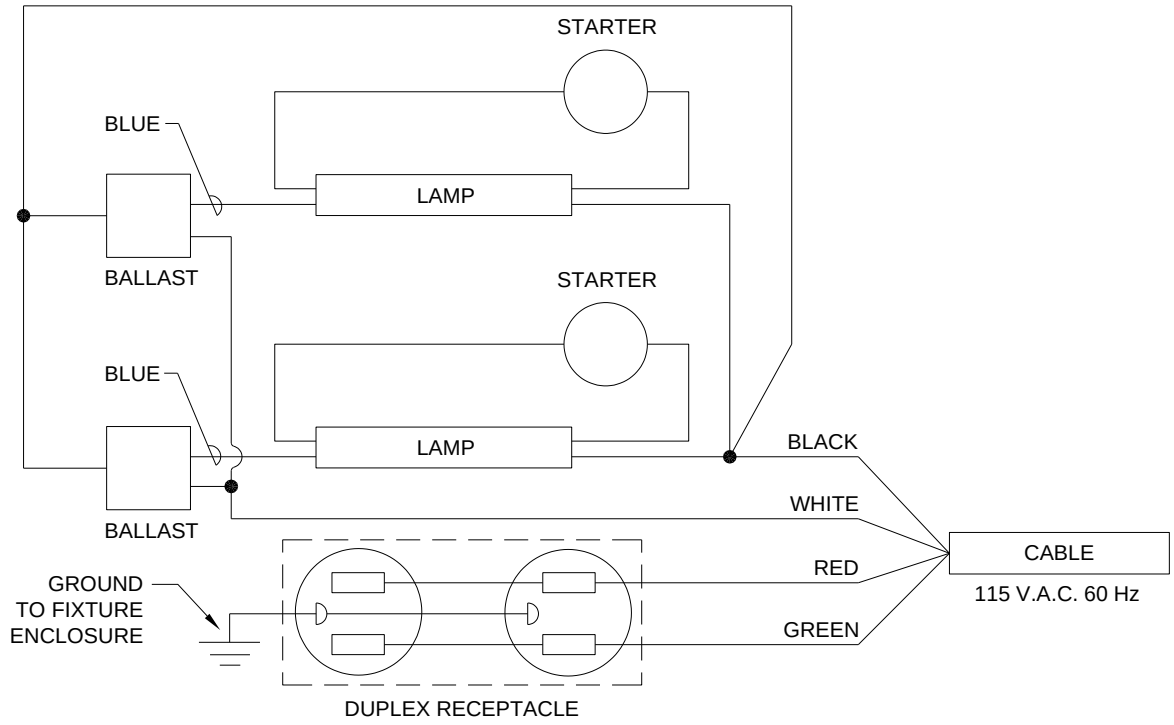
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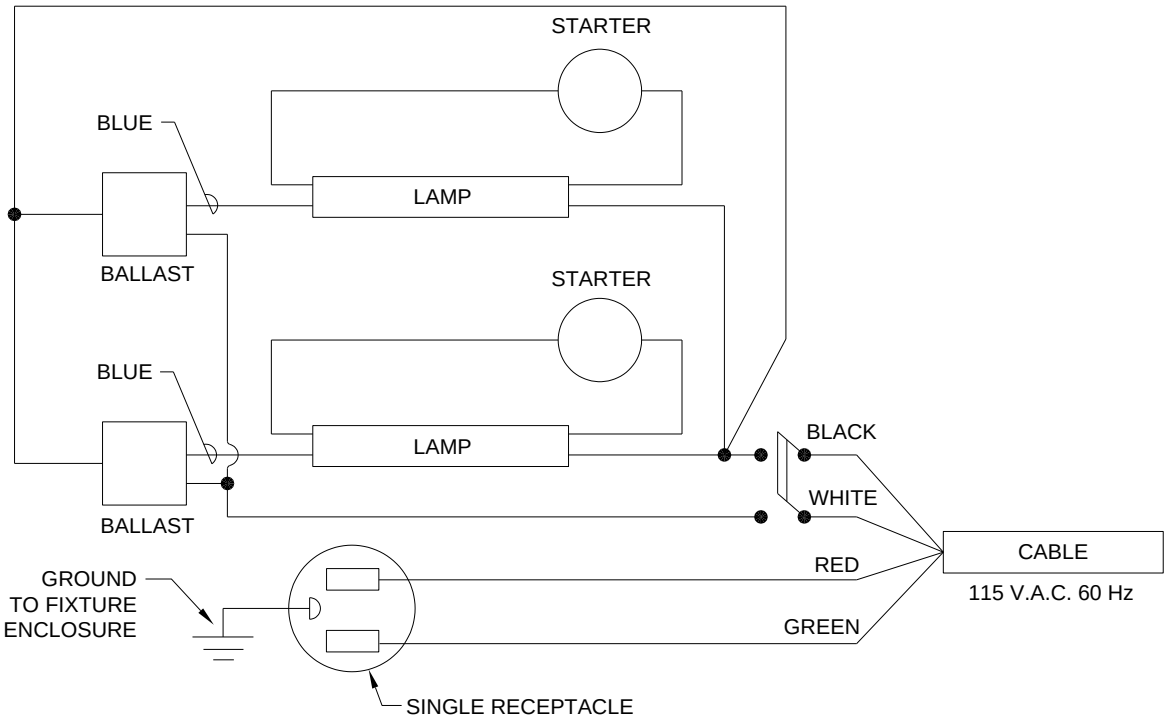
NOTE: Mounting bolts shall be #12 thread-forming screws.

FIGURE 6. Mounting dimensions.

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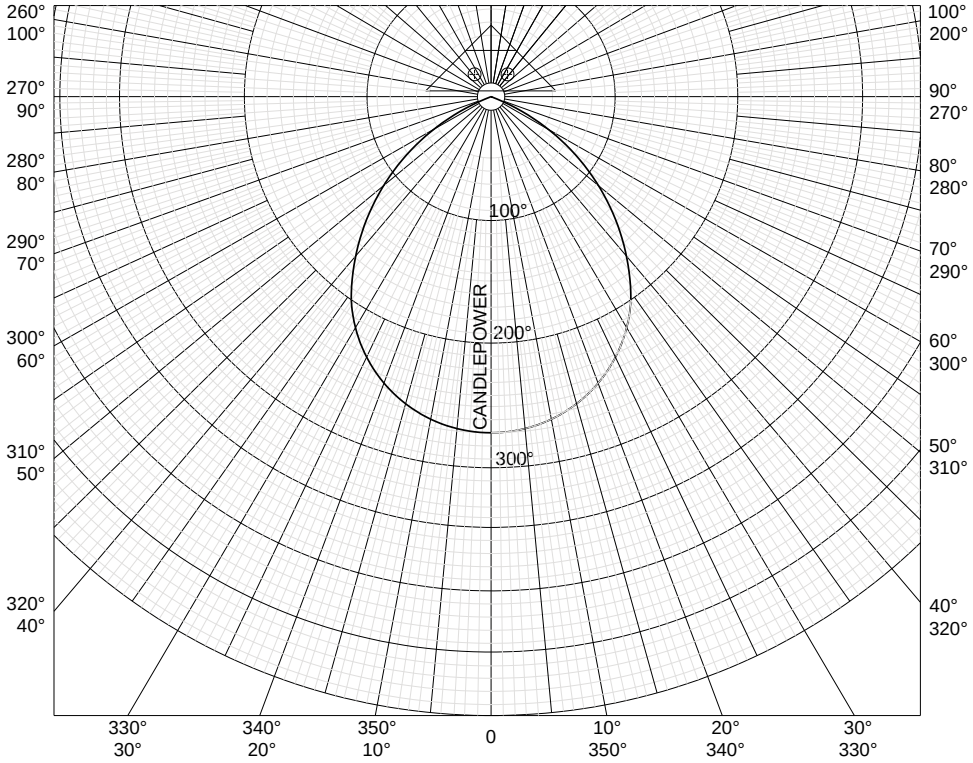
SYMBOLS 351 AND 351.1



SYMBOLS 352 AND 352.1

FIGURE 7. Wiring diagram.

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ANGLE	CANDLEPOWER
	SYMBOLS 351, 351.1, 352, 352.1
0	273
5	270
10	267
15	258
20	250
25	232
30	220
35	202
40	174
45	145
50	116
55	80
60	58

ANGLE	CANDLEPOWER
	SYMBOLS 351, 351.1, 352, 352.1
65	46
70	23
75	12
80	6
85	0
90	0
95	0
100	0
105	0
110	0
115	0
120	0
125	0

Photometric test data:

Distance: 10 feet
Plane: At right angles to long axis of fixture through geometric center of lamp
Voltage: 115 volts, 60 hertz
Ambient temperature: 76 °F
Lamps: Use lamps furnished with fixtures seasoned for at least 100 hours
Ballast: Use ballast furnished with fixture
Lamp correction: Distribution curves based on 790 lumens per lamp

FIGURE 8. Candlepower distribution curves.

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FLUORESCENT REQUIREMENTS

CLASSIFICATION

Type: I
Class: 1

SYMBOL NUMBERS

351: Complete fixture, 24 inches in length, with two 15-watt cool white fluorescent lamps, clear prismatic window, and duplex receptacle.
351.1: Same as symbol 351, except 21 inches in length.
352: Complete fixture, 21 inches in length, with two 15-watt cool white fluorescent lamps, clear prismatic window, and with toggle switch and single receptacle.
352.1: Same as symbol 352, except 24 inches in length.

DESIGN

Electrical rating:	30 watts, 120 volts, 60 hertz.
Dimensions and configuration:	See figures 2 through 4.
Window assembly:	See figure 5.
Mounting diagram:	See figure 6.
Wiring diagram:	See figure 7.
Distribution curves:	Candlepower – see figure 8.
Cable:	10 feet of type LSFHOF-3 in accordance with MIL-DTL-24643/3.
Receptacle for symbols 351 and 351.1:	Receptacle shall be in accordance with W-C-596/12 and National Electrical Manufacturers Association (NEMA) WD-1 (Industry No. 5262), except that the ground contact and mounting strap shall be nickel-plated.
Receptacle for symbols 352 and 352.1:	Receptacle shall be in accordance with Industry No. 5284, except that the ground contact and mounting strap of receptacle shall be nickel-plated. A "U"-shaped electrical insulation paper (fish paper, 0.015-thick, NEMA VU) shall be placed around the receptacles to isolate the electrical connections from the light housing.
Finish:	Complete fixtures shall be finished the same as for type I, class 2 fixtures of MIL-DTL-16377.
Switch for symbols 352 and 352.1:	Switch shall be toggle, double-pole, single-throw (DPST), in accordance with MS16569-1.
Enclosure:	Dripproof, 15 degrees.
Weight:	
Symbols 351 and 352.1:	12 pounds, 6 ounces maximum.
Symbols 351.1 and 352:	11 pounds, 14 ounces maximum.
Designation and marking:	The following labels shall be furnished in accordance with MIL-DTL-16377, applied as shown on figure 4: (a) Two type I. (b) One type IV. The information to be shown on the type IV label shall be: "USE THE FOLLOWING REPAIR PARTS ONLY." List all standard parts shown in table I and window assemblies of table IV by name and stock number only.
Standard parts:	The standard parts shown in table I shall be used in the construction of the fixtures specified herein.

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SOLID-STATE LIGHTING (SSL) REQUIREMENTS

CLASSIFICATION

Type: III
Class: 1

SYMBOL NUMBERS

351SSL: Complete solid-state fixture, 24 inches in length, clear prismatic and duplex receptacle.
351.1SSL: Same as symbol 351SSL, except 21 inches in length.
352SSL: Complete solid-state fixture, 21 inches in length, clear prismatic and with toggle switch and single receptacle.
352.1SSL: Same as symbol 352SSL, except 24 inches in length.

DESIGN

Electrical rating:	15 watts maximum, 120 volts, 60 hertz.
Dimensions and configuration:	See figures 2 through 4.
Distribution curves:	Candlepower – see figure 8.
Cable:	10 feet of type LSFHOF-3 in accordance with MIL-DTL-24643/3.
Receptacle for symbols 351SSL and 351.1SSL:	Receptacle shall be in accordance with W-C-596/12 and NEMA WD-1 (Industry No. 5262), except that the ground contact and mounting strap shall be nickel-plated.
Receptacle for symbols 352SSL and 352.1SSL:	Receptacle shall be in accordance with Industry No. 5284, except that the ground contact and mounting strap of receptacle shall be nickel-plated.
Switch for symbols 352SSL and 352.1SSL:	Switch shall be toggle, DPST, in accordance with MS16569-1.
Enclosure:	Dripproof, 15 degrees.
Chromaticity:	See 3.7.4 of MIL-DTL-16377.
Fail-safe circuit:	See 3.17 of MIL-DTL-16377.
Wiring:	Wiring shall maintain same control scheme as legacy fluorescent fixture.
Weight:	
Symbols 351SSL and 352.1SSL:	12 pounds, 6 ounces maximum.
Symbols 351.1SSL and 352SSL:	11 pounds, 14 ounces maximum.
Designation and marking:	One type I label shall be furnished in accordance with MIL-DTL-16377.

EXAMINATIONS AND TESTS

Examinations and tests shall be as specified in MIL-DTL-16377 and as follows:

First article inspection:	
Complete fixtures:	See table II for type I and table IV for type III.
Window assemblies:	See table III.
Conformance inspection:	
Comparison inspection:	Same as the first article inspection.
Conformance inspection:	See MIL-DTL-16377.

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GENERAL INFORMATION

Assignment of military part numbers and general information is as shown in table V.

TABLE I. Standard parts.

Name	Specification or Military Part Number	National Stock Number (NSN)	Remarks
Lamp	W-L-116/13	6240-00-583-3668	Industry number F15T12/CW
Lamp lock	M16377/40-002	6250-00-399-7040	---
Lampholder	UL 496	6250-00-284-0481	Supplement SB, medium bipin
Startersocket	UL 496	6250-00-295-2738	Supplement SB, medium bipin starter socket
Starter	20W, ANSI C78.180, type III	6250-00-299-2884	Commercial designation: FS2
Ballast	M16377/72-001	6250-00-569-9502	---
Switch, toggle, DPST	MS16569-1	5930-00-719-9659	---

TABLE II. First article and comparison inspection for complete type I fixtures.

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.12	4.5	---
Operation	3.12	4.8.1	---
Dielectric withstanding voltage	3.6.15	4.8.2	---
Insulation resistance	3.6.16	4.8.3	---
Enclosure effectiveness	3.5.11	4.8.14.2	Dripproof, 15 degrees
Shock	3.5.12	4.8.7	Grade B
Vibration	3.5.13	4.8.8	---
Enclosure effectiveness	3.5.11	4.8.14.2	Dripproof, 15 degrees
Dielectric withstanding voltage	3.6.15	4.8.2	---
Insulation resistance	3.6.16	4.8.3	---
Noise	3.5.14	4.8.9	---
Continuity of grounding	3.6.13.4	4.8.12	---
Optical uniformity	3.7.4.1.1	4.8.4	---
Light output	3.7.1	4.8.6	Candlepower
Magnetic permeability	3.4.8	4.8.18	---
Grounding circuit	3.6.13.3	4.8.11	---

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TABLE III. First article and comparison inspection for window assemblies.

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377
Examination	3.13	4.5
Optical uniformity	3.7.4.1.1	4.8.4

TABLE IV. First article and comparison inspection for type III luminaires.

Test	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.12	4.5	---
Operation	3.12	4.8.1	---
Power interface	3.6.18	4.8.21	---
Electromagnetic interference	3.6.14	4.8.16	---
Dielectric withstanding voltage	3.6.15	4.8.2	1/
Insulation resistance	3.6.16	4.8.3	1/
Enclosure effectiveness	3.5.11	4.8.14	Dripproof, 15 degrees
Shock	3.5.12	4.8.7	Grade B
Vibration	3.5.13	4.8.8	1/
Enclosure effectiveness	3.5.11	4.8.14	Dripproof, 15 degrees
Dielectric withstanding voltage	3.6.15	4.8.2	1/
Insulation resistance	3.6.14	4.8.3	1/
Noise	3.5.14	4.8.9	1/
Continuity of grounding	3.6.13.4	4.8.12	---
Photometrics	3.7.1	4.8.6	Candlepower
Brightness	3.7.5	4.8.6.4	---
Chromaticity	3.7.4	4.8.6.5	---
Correlated color temperature	3.7.4.1.1	4.8.6.6	---
Lumen maintenance	3.7.11	4.8.20	---
Fail-safe circuit design	3.17	4.8.24	---
Optical uniformity	3.7.7.1	4.8.4	---
Stress relief	3.8.5.3	4.8.5	---
Magnetic permeability	3.4.8	4.8.18	---
Grounding circuit	3.6.13.3	4.8.11	---

NOTE:

1/ In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.

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TABLE III. Part number and general information.

Military Part Number M16377/58-	Item Name	National Stock Number (NSN)
351	Complete fixture, symbol 351	6210-00-115-9171
351.1	Complete fixture, symbol 351.1	6210-01-044-5273
352	Complete fixture, symbol 352	6210-00-115-9137
352.1	Complete fixture, symbol 352.1	6210-01-045-7724
001	Window assembly, symbols 351 and 352.1	6210-00-115-9197
002	Window assembly, symbols 351.1 and 352	---
351SSL	Complete fixture, symbol 351SSL	---
351.1SSL	Complete fixture, symbol 351.1SSL	---
352SSL	Complete fixture, symbol 352SSL	---
352.1SSL	Complete fixture, symbol 352.1SSL	---

CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 6210-2023-016)

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